

Hey, everybody. Welcome to your chapter 6 review. As you know, the human body is made of many different systems and subsystems that work collectively to support normal physiological functioning. These distinct yet interdependent systems also influence and impact the decisions you will make as a fitness professional to improve your clients' health and fitness. In this episode, we're going to review the cardiorespiratory, endocrine, and digestive systems.

By the end of this chapter, you should be able to identify the integrated structure and function of the cardiorespiratory system, categorize the basic structure and function of the endocrine system, and summarize the basic structure and function of the digestive system. Let's jump right in. The cardiorespiratory system is the umbrella for the cardiovascular and respiratory systems. The cardiovascular system includes the heart, blood, and blood vessels. This system is responsible for circulating and transporting blood throughout the body.

The respiratory system is comprised of the airways and the lungs. The ability to breathe is driven by a functioning respiratory system. Oxygen that we inhale is used to oxygenate our blood. And the unnecessary carbon dioxide is expelled from the body when we exhale. These systems work together to provide the body with adequate oxygen, deliver nutrients, and as a means to remove waste products from the cells in the body.

The heart is a muscular pump which contracts rhythmically to regularly pump blood throughout the body. It is divided into four hollow chambers, two atria, which gather blood returning to the heart, and two ventricles on each side, which pump blood away from the heart to the lungs and body. The right side of the heart is referred to as the pulmonic side. This side of the heart receives blood that is deoxygenated and pumps it into the lungs to be saturated with oxygen.

The left side of the heart is referred to as the systemic side. It receives the oxygenated blood from the lungs and pumps it out to the rest of the body. Each chamber of the heart is separated from one another via one-way valves to prevent backflow of blood. The SA node, located in the right atrium, is referred to as the pacemaker of the heart because it initiates the electrical signal that causes the heart to beat. The internodal pathways transfer the impulse from the SA node to the atrioventricular AV node. The AV node is located between the atria and ventricles and delays the impulse before allowing it to move on to the ventricles.

Heart rate and the stroke volume make up the overall performance of the heart. A normal heart rate or pulse ranges from 60 to 100 beats per minute, and the body will increase its heart rate in response to exercise and decrease the heart rate during sleep. Stroke volume is ultimately a product of end systolic volume, which is the amount of blood present in the ventricle after contraction minus end diastolic volume, or the volume of blood in the ventricles prior to contraction.

Cardiac output is the volume of blood pumped out of the heart in a minute and is a function of both heart rate and stroke volume. Another way to think of it is a combination of the number of times the heartbeats per minute and the amount of blood being pumped out with each beat. Don't forget, normal blood pressure is a systolic reading less than 120 millimeters of mercury, with diastolic less than 80 millimeters of mercury. Individuals with a systolic blood pressure of greater than 139 and a diastolic pressure greater than 89 are classified as being hypertensive.

Now, people may have varying degrees of enthusiasm about blood, but we can all agree it's important for survival. Blood supplies the body's organs and cells with oxygen and nutrients and helps to regulate body temperature, fight infections, and remove waste products. Blood is made up of cells within plasma, a nutrient-rich, watery substance. The three types of cells in our blood are red blood cells, white blood cells, and platelets. They each have their own role.

Red carries the oxygenated blood through the body. White fights infection. And platelets help with clotting. Blood is transported away from the heart through arteries, which have smaller components called arterioles. The vessels that return blood to the heart are called veins, which have smaller components called venules. Capillaries are the smallest blood vessels and connect venules with arterioles and function as an exchange channel between the vessels and bodily tissues. As blood travels close to the skin, it can be either heated or cooled, depending on the external environment.

Let's take a breath and get into the respiratory system.

[TAKES DEEP BREATH]

The respiratory system's job is to collect oxygen from the environment and transport it to the body. As cells use oxygen, they produce carbon dioxide, which is transported back to the heart and lungs in the deoxygenated blood to be released through exhalation. The respiratory system consists of airways, lungs, and respiratory muscles. Its primary function is to work closely with the cardiovascular system to ensure proper cellular function, another amazing process that we often take for granted while it's just happening.

I'll bet you're hyper-aware of your breathing right now, aren't you? Breathing-- or ventilation, as it's also called-- is divided into two phases, inspiration and expiration, the Inhale and the exhale. We all know how important the use of breath is. But why? Well, breathing has many functions that help the body regulate itself and work more effectively. Inspiratory ventilation is active, and expiratory ventilation can be both active and passive.

During normal breathing, it's passive, resulting from the relaxation of the contracting inspiratory muscles, the diaphragm, and the intercostals. During forced or heavy breathing, expiratory ventilation actually relies on secondary muscles to compress the thoracic cavity and force air out. Holding in breath during exercise while contracting the abdominals increases the rigidity of the spine to make it easier to support heavy loads, also known as the Valsalva maneuver.

So you can see how our musculoskeletal system is involved as well. There are two groups of respiratory passages, the conducting airways and the respiratory airways. The conducting airways consist of all the structures through which air travels before entering the respiratory airways. These structures purify, humidify, and warm or cool air to match body temperature. The respiratory airways collect the channeled air coming from the conducting airways and allow gases like oxygen and carbon dioxide to be transferred in and out of the bloodstream in a process called diffusion.

Another term you should know is oxygen consumption. That is the actual usage of oxygen by the body. Maximal oxygen consumption is the highest rate of oxygen transportation and utilization achieved at maximal physical exertion. A normal respiratory rate is 12 to 16 breaths per minute. A respiratory rate of less than 8 breaths per minute would be considered too slow or what we call bradypnea. A rate of greater than 24 breaths per minute is considered too high or tachypnea.

Alterations in breathing patterns can have a direct negative impact on the human body. Dysfunction such as labored breathing patterns will undoubtedly affect exercise performance. If the breathing patterns become shallow and the body uses those secondary respiratory muscles more than the diaphragm, it can be hard to maintain correct posture, which may create excessive muscular tension in other areas. When breathing gets out of whack, it can also lead to headaches, lightheadedness, and dizziness.

Short, shallow breaths can even lead to altered carbon dioxide and oxygen blood content, causing feelings of anxiety. Inadequate oxygen and retention of metabolic waste within muscles can also create stiff muscles and joints. If you have a client that's complaining of headaches, anxiety, fatigue, poor sleep patterns, or poor circulation-- if you have a client that's complaining of headaches, anxiety, fatigue, poor sleep patterns, or poor circulation. It's best to refer them to a medical professional to get checked out right away.

On to the endocrine system-- here, we'll find the glands that secrete hormones responsible for physiological functioning as well as growth and development. The endocrine system is a system that regulates a variety of bodily functions, including the control of mood, growth and development, tissue function, and metabolism. It consists of glands, hormones, or chemical messengers, and target receptor cells.

When these chemical messengers are released into the bloodstream, they are protected by personal security guard-like transporters, which carry them to the intended organ or structure, where they bind with the receptor to stimulate a particular function. The primary endocrine glands are the hypothalamus, pineal, pancreas, thyroid, pituitary, adrenal, and reproductive glands.

Control of blood glucose is regulated in part by insulin and glucagon, which are secreted by the pancreas. They work opposite one another to control blood glucose levels. Glucagon aids in the metabolism of glucose. And insulin aids in the cellular uptake and storage of glucose. As exercise levels increase, glucose uptake by the body's cells will also increase as a result of an increased sensitivity of the cells to insulin.

As a result, insulin levels will drop during physical activity, while glucagon secretion by the pancreas increases to help maintain a steady supply of blood glucose. Cortisol, which is stimulated by the adrenal cortex, may be used to aid in recovery from exercise and as a marker of overtraining. Cortisol is considered a catabolic hormone and is sensitive to blood sugar and sleep. Overall, adequate sleep is necessary for glucose metabolism, hormone function, and muscle recovery. So encourage your clients to avoid sleep deprivation for optimum performance.

Next, we can look at the thyroid. Thyroid hormones serve numerous functions in the body, including metabolism and increasing bone mineral density through the secretion of calcitonin. Moving on to the catecholamines epinephrine and norepinephrine, they prepare the body for activity and are immediately stimulated from the adrenal medulla in response to exercise. Testosterone levels can also be stimulated through intense exercise, even though levels decline with age.

Growth hormones, which are responsible for growth and development as well as lipolysis, are produced from the pituitary gland. One of the most potent of the anabolic hormones is insulin-like growth factor, which is produced by the liver in response to growth hormones binding on liver receptors. Testosterone, growth hormones, and insulin-like growth factors are stimulated in response to anaerobic resistance training as well as vigorous aerobic activity.

Bear in mind that physical activity has a significant impact on hormone secretion. As a CPT, if you are familiar with hormone response to exercise, you can maximize programming strategies and avoid overtraining. One last system to digest-- I bet you can see where this is going. The food and drink we take in rely on the digestive system to break down and send the right materials to the right places in our bodies or out of our bodies.

The digestive system consists of the oral cavity, the upper GI system, the lower GI tract, as well as the liver, gallbladder, and pancreas. Digestion starts in the oral cavity, where materials are broken down through mastication, aka chewing with help from our tongue, teeth, saliva, and enzymes in the mouth. Then food and liquids pass into the esophagus and eventually into the stomach, where gastric juices aid in digestion, kill bacteria, and turn food into chyme, which is then passed into the small intestine.

Both the small intestine and large intestine are responsible for absorption of many essential nutrients. The liver, gallbladder, and pancreas aid in this process by producing and storing digestive juices, which are secreted into the small intestine to help with the digestion. Ultimately, what's not absorbed and used or stored as energy source is eliminated as waste. Digestion is complex and requires the cooperation of both the upper and lower GI systems. These systems function via stimulation from our specific diets, fluid intake, medications, and activity levels.

Inadequate dietary intake, like not getting enough fiber or hydration, can cause digestive problems. Digestive function also requires integration from the endocrine and neural systems. And although you may think that the digestive system applies more to the nutrition side of things, evidence suggests that exercise can improve digestive function by increasing transit time of food from the upper to the lower GI tracts and help with constipation.

It's pretty clear that the cardiovascular, respiratory, endocrine, and digestive systems greatly influence a client's health and fitness levels. The greater your expertise on the functionality of the body and how exercise impacts the programs you design, the more you will set yourself apart as a true leader in the industry. So with that, I'll leave you to it. Be well, and happy studying.